

PROJECT PROPOSAL

BLOCK 1D-2025-2026

Day to day stock price predictor

Name of the Group

Group 15

Names of the Student

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Date

12/05/2026

I. Introduction

The project focuses on developing a binary classification machine learning tool that predicts whether **Microsoft (MSFT)** stock will go **up** or **down** in the next trading day, displayed in an interactive dashboard.

The tool utilizes **exogenous** data (such as crude oil, gold, etc.) combined with MSFT stock data via statistical modeling and machine learning. The stakeholders are team members, academic supervisors, and retail investors, the last being end user and main stakeholder. This is essential to develop such a tool, as non-professional investors would significantly benefit from it.

It is possible to build it, because there is **enough financial data** across Internet, and more importantly, such tools already exist. They benefit from applying machine learning to spot **non-linear** stock market behaviour and help users analyze market better.

Additionally, applying machine learning is valuable, because as mentioned above, it can identify non-linear patterns a human is very unlikely to. The project utilizes exogenous data, as well as MSFT stock records. That is difficult for a human, but not for a machine learning algorithm. On top of that, with the use of explainable AI techniques, it is possible to **spot** the main elements that affect a stock's likelihood of going up or down.

II. Problem Statement

Predicting short-term stock price movements is one of the most challenging problems in quantitative finance. The prediction and forecasting of asset prices and returns remains one of the most challenging problems for quantitative finance and practitioners alike, and the massive increase in data generated in recent years presents an opportunity to leverage machine learning algorithms (Ndikum, 2020). For investors and portfolio managers, even a small improvement in forecast accuracy can translate directly into better decision-making, reduced risk, and improved returns.

This project focuses on Microsoft Corporation (MSFT), one of the most actively traded stocks on the Nasdaq-100. The core problem is: given historical price data and relevant external signals, can we reliably predict whether the next-day closing price of MSFT will go up or down? Beyond the model itself, the solution must be accessible to end users through a functional application. A standalone prediction model has limited value without an interface that allows users to interpret and act on its predictions.

Solving this problem has clear business value. Machine learning algorithms use past data to create short-term forecasts of stock price movements, and models such as Random Forest and neural networks have shown strong accuracy across evaluation metrics including Accuracy and F1 Score (Albert Wong, 2023). Accurate next-day closing price direction predictions can support trading strategies, help risk managers set better stop-loss levels, and give financial analysts a data-driven basis for their recommendations.

The goals of this project are:

- Develop a machine learning model that predicts whether the next-day closing price of MSFT will go up or down
- Incorporate both historical price features and external sentiment signals to improve accuracy
- Evaluate model performance using appropriate classification metrics including Accuracy, Precision, Recall and F1 Score
- Build a user-facing application that presents predictions, visualisations, and insights in an accessible way
- Present predictions through an interactive dashboard accessible to both technical and non-technical users

III. Data Description

Data Sources

This project will use four datasets sourced from the Nasdaq-100 stock market data repository. All datasets cover a 10-year period from February 2015 to February 2025.

MSFT Daily Stock Data The primary dataset contains daily trading records for Microsoft Corporation (MSFT), one of the most actively traded stocks on the Nasdaq-100. It contains daily OHLCV data — Open, High, Low, Close, and Volume — which will serve as the core features for the prediction model. This dataset will form the foundation of the project, providing the historical price behaviour the model will learn from.

Gold Prices Daily closing prices for gold over the same 10-year period. Gold will be included as an exogenous feature because it serves as a safe-haven asset — when market uncertainty rises, investors tend to move money into gold, which historically correlates negatively with stock prices. Only the daily closing price will be used.

Crude Oil Prices Daily closing prices for crude oil over the same 10-year period. Crude oil will be included because energy costs directly affect the profitability of large technology companies and influence broader market sentiment. Only the daily closing price will be used.

VIX — Market Volatility Index The VIX index measures expected market volatility and is widely known as the market's fear gauge. A high VIX indicates investor uncertainty and typically correlates with stock price declines. The VIX will be extracted from the Nasdaq-100 macroeconomic indicators dataset. Other indicators in this file such as inflation and interest rates will be excluded due to their monthly update frequency being unsuitable for daily prediction tasks.

Dataset Structure

After merging all four sources on date, each row in the combined dataset will represent one trading day and contain the following features:

date	All	Trading date
open	MSFT	Opening price
high	MSFT	Daily high price
low	MSFT	Daily low price
close	MSFT	Closing price
volume	MSFT	Number of shares traded
gold_close	Gold	Gold closing price (USD)
oil_close	Crude Oil	Crude oil closing price (USD)
vix	Macro	VIX market volatility index

Potential Data Quality Issues and Handling

Missing Values Since the four datasets come from different sources, minor misalignments in trading calendars are expected — for example, MSFT trading days may not always have a corresponding gold or VIX entry. These gaps will be handled using forward fill, which carries the last known value forward. This is the standard approach for financial time series data and will ensure no rows are lost due to minor calendar differences.

Date Format Inconsistencies The datasets are expected to contain minor formatting differences in the date column, such as timestamps or timezone information. These will be cleaned during the data loading stage to ensure consistent date formatting across all sources before merging.

Neutral Days After computing next-day returns to create the prediction target, days where the price movement falls within a small threshold will be considered neutral and excluded from the dataset. This is because very small price moves carry little predictive signal and would add noise to the model.

Outliers Stock price data naturally contains extreme values during significant market events such as financial crises or earnings surprises. These extreme values represent genuine market behaviour rather than data errors, so no outlier removal will be applied. Their influence will be managed through feature scaling during preprocessing.

IV. Methodology

4.1 Data Collection and Preprocessing

All CSV files will be loaded into a SQLite database, which serves as the central data storage solution for the entire project. The MSFT date range serves as the reference index to which all other datasets are aligned. Missing values caused by minor trading calendar misalignments are handled using forward fill, carrying the last known value forward — the standard approach for financial time series data.

Engineered features include lag features (previous 1 and 2 day closing prices), rolling statistics (5-day and 10-day moving averages), daily return percentage, price range (daily high minus low), and daily return percentages for gold and crude oil. All features will be scaled using RobustScaler fitted on training data only to prevent data leakage. RobustScaler uses the median and interquartile range rather than the mean and standard deviation, which reduces the influence of extreme observations such as the March 2020 COVID-19 swings on the scaling parameters.

4.2 Model Selection and Training

The task is a supervised binary classification problem: predict whether the next-day closing price of MSFT will go up or down based on features available up to the current trading day t , predicting the direction at day $t+1$. A random baseline of approximately 50% accuracy is expected from random guessing given the binary nature of the task. All subsequent models must outperform this baseline.

Four machine learning models will be developed iteratively, each building on the findings of the previous one. Iteration 1 uses Logistic Regression as a linear baseline. Iteration 2 uses Random Forest for non-linear pattern detection and feature importance analysis. Iteration 3 uses XGBoost with hyperparameter tuning, expected to be the strongest performing traditional model. Iteration 4 uses an LSTM neural network to capture long-range temporal dependencies in the price history.

As a final iteration, the best-performing base models from iterations 2 through 4 (Random Forest, XGBoost, and LSTM) are combined into a stacking ensemble. A linear meta-learner is trained on out-of-fold predictions to learn the optimal weighting of each model's contribution. Random Forest contributes non-linear capacity and resistance to noisy features, XGBoost adds sequential error-correction that captures subtle patterns the others miss, and LSTM provides temporal awareness of long-range dependencies. When one model underperforms in a given market regime, the others compensate, producing more consistent predictions across different conditions than any single model alone.

A time-based split is used to prevent lookahead bias: 80% training (February 2015 to February 2023) and 20% testing (February 2023 to February 2025). No shuffling is applied to preserve the temporal structure of the time series.

4.3 Evaluation Metrics

Since this is a binary classification problem, the following metrics will be used to evaluate model performance. Accuracy measures the overall percentage of correct UP and DOWN predictions. Precision measures of all days predicted as UP how many were actually UP. Recall measures of all actual UP days how many the model correctly identified. F1 Score is the harmonic mean of precision and recall and serves as the primary evaluation metric, balancing both false positives and false negatives. A Confusion Matrix provides a visual breakdown of correct and incorrect predictions for both classes. All models are benchmarked against the 50% random baseline.

4.4 Deployment

The final stacking ensemble will be deployed within a Streamlit dashboard connected to the SQLite database. The dashboard reads predictions directly from the predictions table, which is populated by the model after each run. This ensures the dashboard always displays the latest results without requiring any manual intervention. The application will be containerised using Docker and Docker Compose, making it fully portable across different machines. A README with Poetry lock file will ensure reproducibility.

V. Project Timeline

Week	Phase	Key tasks	Deliverables	Deadline
Week 4	Application architecture + final proposal	ML modelling (Task 10), app architecture (Task 13), production ready code (Task 11), group proposal (Task 6)	Group proposal document, model notebook	Tuesday: final proposal presentation
Week 5	Data engineering	Data pipeline (Task 9), database setup (Task 8),	Pipeline scripts, database scripts,	Friday: retrospective

		model iteration (Task 10), dashboard start (Task 13)	processed dataset	
Week 6	Containerisation + design report	Docker (Task 13), design report (Task 7), legal compliance (Task 4.2)	Docker files, design report, legal report	Friday: design report
Week 7	Integration + testing	Unit tests + 80% coverage (Task 12), dashboard integration (Task 13), model finetuning (Task 10), README (Task 12)	Tested application, coverage report, README	Friday: retrospective
Week 8	Finalise + present	Final presentation (Task 14), peer feedback (Task 3), finalise all deliverables	Live demo, presentation media	Thursday: final presentation. Friday: all deliverables

Team contributions:

Task	Responsible
Task 4.2 - Legal compliance report	Gádor, Gergely
Task 6 - Group project proposal	All members
Task 7 - Application design document	Dekker, Lucan den
Task 8 - Database management	Qusaily, Qusai Al
Task 9 - Data processing pipelines	Kambarov, Timur

Task 10 - ML model development	All members
Task 11 - Modular maintainable code	All members
Task 12 - Unit testing + documentation	All members
Task 13 - Application development	Dekker, Lucan den
Task 14 - Final presentation	All members

VI. Reference

Albert Wong, S. W.-W. (2023). *Short-Term Stock Price Forecasting using exogenous variables and Machine Learning Algorithms*. Retrieved from Trading and Market Microstructure: <https://arxiv.org/abs/2309.00618>

Ndikum, P. (2020). *Machine Learning Algorithms for Financial Asset Price Forecasting*. Retrieved from arxiv: <https://arxiv.org/abs/2004.01504>